Daily Scholar Papers Report — 2026-06-09

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Window covered: 2026-06-08 → 2026-06-09 (Google Scholar alerts + user-curated self-emails, last 24 h)

Executive Summary

A small, quiet window. The Keep slot goes to a Shengchao-Qin-group preprint that re-frames LLM code-readability steering as a **multitask representation-engineering** problem under multidimensional orthogonal constraints (MOC-JPCA), with two formal theorems bounding the readability gain (lower) and the correctness loss (upper). The bounded-tradeoff framing is the methodologically interesting part: at tuned coefficient combinations, correctness drops by **0.28 % / 0.79 % / 0.95 %** across Codellama 13b / Deepseek R1 14b / Qwen2.5coder 14b Instruct while readability metrics rise. The Borderline-High bucket holds three AsiaCCS-class entries available abstract-only: a hardware-fuzzing × sanitizer hybrid from the Texas A&M / TU Darmstadt / Intel group (**FUZZItizer**); an LLM-driven REST API security fuzzer from Würzburg / Duisburg-Essen (**RESTing-LLAMA**); and a Zhenchang-Xing-coauthored **SoK on XR Privacy-UX tradeoffs**. No user-curated self-emails arrived; no exclusion-list matches.

Outstanding: 0 · **Keep:** 1 · **Borderline High-Priority:** 3

Highlighted Papers

#	Title	Authors	Venue	Link
2.1	Towards the Readability of LLM-Generated Codes through Multitask Representation Engineering	H. Gao, L. He, Y. Pan, S. Yu, Y. Zeng, S. Qin, W. Sun	arXiv 2606.06214	arXiv
3.1	FUZZItizer: Hardware Sanitizer-Assisted Fuzzing for Automated SoC Vulnerability Detection	R. Kande, M. Rostami, C. Chen, H. Khattri, J.M. Fung, A.-R. Sadeghi, J. Rajendran	AsiaCCS 2026	DOI
3.2	RESTing-LLAMA: Large Language	V. Gadey, C. Sendner, K.	AsiaCCS 2026	DOI

#	Title	Authors	Venue	Link
	Model based REST API Fuzzing	Zimmermann, A. Dmitrienko		
3.3	SoK: Navigating the Privacy-UX Trade-offs in Extended Reality (XR) — A Socio-Technical Taxonomy and Research Roadmap	S. Wang, M.A.P. Chamikara, M. Baruwal Chhetri, Z. Xing, et al.	Proc. ACM 2026	DOI

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2. Keep

2.1 · LLM CODE READABILITY · arXiv 2026 — multitask representation engineering with orthogonality-constrained joint PCA (MOC-JPCA); two theorems bound readability gain (lower) and correctness loss (upper); <1 % correctness loss across three 13–14B models

2.1 Towards the Readability of LLM-Generated Codes through Multitask Representation Engineering — Gao, He, Pan, Yu, Zeng, Qin, Sun (Xiamen U / Shenzhen U / Fujian Normal U / Northumbria U / Xidian U / Peking U), arXiv 2026

Authors: Huifan Gao, Liuhua He, Yinghui Pan, Shenbao Yu, Yifeng Zeng, **Shengchao Qin**, Weidi Sun. **Venue:** arXiv 2606.06214, IEEE journal submission, 4 Jun 2026. Mirrors: PDF · Scholar lookup. **License:** arXiv non-exclusive — no figure embedding; structural diagrams are Mermaid recreations.

Why surfaced today. Followed-researcher track (Shengchao Qin).

Problem. LLM-code research mostly optimises *correctness*; *readability* is multi-dimensional and subjective, and existing work either evaluates readability post-hoc or relies on unstable prompt engineering. Representation-engineering (RepE) steering is a natural fit because it is low-data, low-compute, and inference-only — but prior RepE is single-property. Extracting per-property steering vectors independently via PCA produces *correlated* directions that interfere when injected together for multi-property control.

Approach — MRepE. Three stages:

1. **Data preparation.** For each readability metric $k \in$ {comment density (CD), naming conventions (NC), cyclomatic complexity (CC)}, build M_k contrastive prompt pairs $(x^{+}_{k,i}, x^{-}_{k,i})$ and compute mean-centred hidden-state differences at layer l:

$$\Delta h_{k,i}^{(l)} = h_\theta^{(l)}(x_{k,i}^{+}) - h_\theta^{(l)}(x_{k,i}^{-})$$

2. **Steering vector extraction — MOC-JPCA.** Jointly extract one principal direction per dataset under multidimensional orthogonal constraints ($d_i \perp d_j$ for $i \neq j$). Solved on the Stiefel manifold via Riemannian gradient + Armijo backtracking + QR retraction.

3. **Steering vector injection.** Apply additive perturbation at selected layers:

$$h_{\theta, v^{(l)}}^{(l)} \leftarrow h_{\theta}^{(l)} + v_1^{(l)} + \ldots + v_K^{(l)} \quad \text{(Eq. 2)}$$

where v^{(l)}_k = c_k · d^{(l)}_k, with c_k tuning per-metric strength.

```
flowchart LR
    A[Per-metric paired prompts<br/>CD/NC/CC, x+ vs x-] --> B[Hidden-state diffs Δh<br/>per layer l]
    B --> C[Mean-centred X_k<br/>per metric k]
    C --> D[MOC-JPCA<br/>joint PCA + orthogonal<br/>constraints on Stiefel manifold]
    D --> E[Orthogonal directions<br/>d_1, d_2, d_3]
    E --> F[Coefficient knob c_k<br/>per metric strength]
    F --> G[Additive injection<br/>h ← h + Σ c_k·d_k]
    G --> H[LLM with tunable<br/>readability vs correctness]
```

Mermaid recreation of the MRepE pipeline; original Fig. 2 is not CC-licensed.

Formal results. From the paper:

- **Theorem 2.** Lower-bounds the increase in expected probability of positive answers to readability-related queries after injection, as a function of c_k.
- **Theorem 3.** Upper-bounds the decline in expected probability of correct answers to correctness-related queries, again parametrised by c_k.

Together they convert RepE from a heuristic toolkit into a *coefficient-tunable system with provable worst-case bounds*. The reader can pick c per metric, predict the worst-case correctness drop, and stop tuning when the predicted loss exceeds tolerance.

Evaluation.

- **Datasets:** MBPP-CR (readability-paired) and MBPP-CC (multiple-choice correctness), each 300 samples extended from MBPP.
- **Models:** Deepseek R1 14b · Qwen2.5coder 14b Instruct · Codellama 13b Instruct.
- **Best-found operating points (paper's verbatim Sec. V.D summary, ≤15 words each):**

Model	c1 (CD)	c2 (NC)	c3 (CC)	Correctness loss
Codellama 13b Instruct	5.0	4.5	5.0	0.28 %
Deepseek R1 14b	1.0	4.0	5.0	0.79 %
Qwen2.5coder 14b Instruct	3.5	5.0	3.0	0.95 %

- **Stat sig:** behaviour changes significant at $p < 1.7 \times 10^{-4}$ across model × metric combinations (Table III).
- **Bound tightness:** Fig. 5 surface plots show measured readability gain vs Theorem-2 lower bound — the green (measured) and red (theoretical bound) surfaces track each other across $c \in [0, 5]$.

Why this matters. Three reasons. (i) The orthogonality-constrained joint PCA is a clean fix for the multi-vector interference problem in steering, and the Stiefel-manifold derivation transfers beyond code. (ii) The theorems make RepE a tunable engineering knob with worst-case-loss guarantees, not a heuristic —

exactly what is needed when a downstream user demands "improve property X without breaking property Y by more than ε." (iii) The judgment-task-as-proxy methodology (validated by Fig. 3 correlation with sequence-level NLL) is itself a reusable analysis pattern, because it sidesteps the length-normalisation and attribute-irrelevant-token confounds that plague sequence-level intrinsic-evaluations.

Caveats. Readability proxies (CD, NC, CC) are themselves imperfect — the paper acknowledges this. The formal analysis is on QA-probability proxies, not raw generation, although Fig. 3 establishes empirical correlation. Three models, all 13–14B class — generalisation to frontier-scale models open.

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3. Borderline High-Priority

3.1 · HARDWARE FUZZING · AsiaCCS 2026 — ports software-style sanitizers (ASan/UBSan analogues) into the RTL/SoC verification fuzzing loop; TAMU × TU Darmstadt × Intel; abstract-only

3.1 FUZZItizer: Hardware Sanitizer-Assisted Fuzzing for Automated SoC Vulnerability Detection — Kande, Rostami, Chen, Khattri, Fung, Sadeghi, Rajendran (Texas A&M / TU Darmstadt / Intel), AsiaCCS 2026

Authors: Rahul Kande, Mohamadreza Rostami, Chen Chen, Hareesh Khattri, Jason M. Fung, Ahmad-Reza Sadeghi, Jeyavijayan Rajendran. **Venue:** AsiaCCS 2026, cycle-1 (accepted-papers list) · DOI 10.1145/3779208.3806078 · ACM author-retained, paywalled. **Access:** No public preprint surfaced at posting time. Summary based on alert abstract snippet and the cycle-1 accepted-papers listing.

Picture. Hardware fuzzing has matured (TheHuzz, HyPFuzz, Fuzzerfly Effect from the same group), but bug *detection* in hardware fuzzers still leans on coarse signals — assertion violations, simulator hangs, or post-hoc waveform inspection — which leaves silent vulnerabilities undetected. FUZZItizer's pitch is to import the *sanitizer* idea from software fuzzing (ASan / UBSan / MSan) into the SoC verification loop: instrument the RTL with hardware-level sanitizer probes (memory access bounds, taint analogues, undefined-behaviour analogues), so the fuzzer detects violations *during* simulation rather than relying on post-hoc crash triage. The Intel co-authorship (Khattri, Fung) suggests evaluation on industrial SoC IP.

Why surfaced. Top security venue (AsiaCCS), well-known hardware-fuzzing lineage, and a directly transferable methodology: sanitizer + fuzzer is the canonical software-side recipe, and the question of "what is the right sanitizer abstraction for RTL" is the natural next step.

Open question. What exactly the "hardware sanitizer" abstraction is — a runtime check inserted by an RTL pass, a co-simulation gadget, or a wrapper around existing UVM/assertion infrastructure — determines how reusable the framework is. Revisit when the PDF lands.

3.2 · LLM API FUZZING · AsiaCCS 2026 — LLM-driven REST API security fuzzer; Würzburg × Duisburg-Essen; abstract-only

3.2 RESTing-LLAMA: Large Language Model based REST API Fuzzing — Gadey, Sendner, Zimmermann, Dmitrienko (U Duisburg-Essen / U Würzburg), AsiaCCS 2026

Authors: Varun Gadey, Christoph Sendner, Keven Zimmermann, Alexandra Dmitrienko. **Venue:** AsiaCCS 2026, cycle-1 (accepted-papers list) · DOI 10.1145/3779208.3785383 · ACM paywall. **Access:** No public preprint located. Summary from alert abstract snippet.

Picture. Yet another entry in the rapidly-growing LLM-for-REST-API-fuzzing line (RESTler → RESTLess → LlamaRestTest → RESTing-LLAMA). The Würzburg / Duisburg-Essen affiliation, combined with the AsiaCCS placement, suggests focus on *security* fuzzing (auth bypass, parameter pollution, IDORs) rather than purely functional bug-finding. Top-venue acceptance earns it a Borderline-High slot even without full text.

Why noted but not Keep. The LLM-API-fuzzing space is now crowded; a real differentiator would be (a) head-to-head against prior LLM baselines on the same benchmark, (b) CVE disclosures, or (c) a methodological insight about *prompt strategy* for security-oriented (vs functional) fuzzing. Defer the deep read until the PDF is available.

3.3 · XR PRIVACY SOK · Proc. ACM 2026 — socio-technical taxonomy of XR privacy-vs-immersion tradeoffs; followed-researcher track (Zhenchang Xing); abstract-only

3.3 [SoK: Navigating the Privacy-UX Trade-offs in Extended Reality (XR) — A Socio-Technical Taxonomy and Research Roadmap]() — Wang, Chamikara, Baruwal Chhetri, Xing, et al., Proc. ACM 2026

Authors: Shaobo Wang, M.A.P. Chamikara, Mohan Baruwal Chhetri, **Zhenchang Xing**, et al. **Venue:** Proceedings of the ACM 2026, DOI [10.1145/3779208.3807495](). Mirrors: [Scholar lookup](). **Access:** ACM paywall; abstract-only from alert snippet.

Picture. SoK structuring privacy-versus-UX tradeoffs in XR. Extended Reality integrates physical and digital environments through continuous multimodal sensing, creating tension between privacy protection and user experience: strong privacy controls (face/iris anonymisation, biometric stripping, sensor gating) protect users but disrupt immersion; weak controls preserve immersion but leak continuous multimodal sensor data. The paper builds a socio-technical taxonomy and a research roadmap.

Why surfaced. Followed-researcher track (Zhenchang Xing). Adjacent rather than central to the code-and-security thread — kept on the radar for the taxonomy framing rather than for the technical content.

Cross-Paper Synthesis

Two visible threads.

Hardware-software fuzzing convergence. FUZZItizer ports the sanitizer idiom from ASan/UBSan into RTL/SoC verification — completing a years-long migration. The classical software-fuzzing playbook (coverage-guided + sanitizer) is now the de-facto template for hardware fuzzing, with the open question being *which* sanitizer abstractions are tractable at gate/RTL level. RESTing-LLAMA, by contrast, occupies the protocol/API frontier where LLMs replace hand-written grammars. Both papers, both AsiaCCS cycle-1: a quiet snapshot of where the non-ML-vs-protocol-fuzzing centre of mass is now (hardware-down and LLM-up).

Property-level decoding-time control with bounded loss. The Qin-group readability paper makes a small but instructive move: representation engineering, once a single-property steering trick, becomes a *multi-property and provably bounded* control mechanism. The Stiefel-manifold MOC-JPCA construction should generalise to any setting where the goal is to steer several conflicting properties with worst-case loss bounds — safety + helpfulness + correctness for chat models; readability + security + performance for code generators.

Writing & Rationale Insights

The Qin paper executes two stylistic moves worth copying. First, *the bounded-loss framing*: instead of pitching “we improved metric M without breaking metric M′,” the paper writes “Theorem 3 upper-bounds the M′ loss; here are coefficient combinations where the bound holds with <1 % observed loss.” This converts an empirical claim into a tunable guarantee — much harder to dismiss. Second, *separating judgment-task probability from generation behaviour*: rather than claim the steering vectors *cause* better generation, the paper uses positive-answer probability on QA queries as a tractable proxy, validates the correlation empirically (Fig. 3), and confines the formal analysis to the proxy. This is a cleaner way to handle the inferential gap between intrinsic-probe results and downstream behaviour than the usual “we improve metric X therefore the model is better” rhetoric.

The two AsiaCCS abstract-only entries are reminders that when the full text is unavailable the right move is brevity and honesty: surface the venue/group signal, summarise the snippet, flag what is missing, and defer the deep read.